15 September 1976

MILITARY SPECIFICATION

ROPE, FIBROUS, PLAITED, POLYESTER/POLYPROPYLENE DUAL FIBER

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers eight strand, plaited, polyester/polypropylene dual fiber rope.

1.2 Classification. The rope shall be furnished in one type in the circumference as specified (see table I and 6.2).

2. APPLICABLE DOCUMENTS

2.1 Issues of documents. The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

UU-T-81 - Tags, Shipping and Stock

MILITARY

MIL-C-3131 - Cordage; Preparation for Delivery of

STANDARDS

FEDERAL

FED-STD-191 - Textile Test Methods

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: U.S. Army Natick Research and Development Command, Natick, MA 01760 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

FSC 4020

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T23:09Z
Check the source to verify that this is the current version before use.

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes

(Copies of specifications, standards and drawings required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 First article. When specified (see 6.2), the contractor shall furnish a sample for first article inspection and approval (see 4.3 and 6.4).

3.2 Government or contractor purchases. The requirements specified in 3.8, 3.9 and 3.10 apply only to rope purchased directly by the Government. All other requirements apply to rope purchased as a component for an end item by a contractor and to rope purchased directly by the Government.

3.3 Materials.

3.3.1 Polyester. The polyester portion of the rope shall be bright, virgin, continuous filament polyester fiber of suitable tenacity to meet the requirements of this specification.

3.3.2 Polypropylene.

3.3.2.1 Cover yarn core. The polypropylene for the core of cover yarns shall be virgin, continuous, monofilament fibers of suitable strength to meet all the requirements of this specification, and shall have a minimum softening point of 300°F and a specific gravity no greater than 0.91. The fibers shall range in size from 100 to 400 denier per filament and shall contain heat and ultraviolet stabilizers.

3.3.2.2 Inside yarns. Inside yarns, when used in construction of the strand (see 3.4.2), shall be virgin, continuous, monofilament polyproplyene yarns or continuous slit or fibrillated film polypropylene. The polypropylene shall contain heat stabilizers. Ultraviolet stabilizers may be used but are not required.

3.4 Construction. The ropes shall be made of eight strands arranged in four pairs. The alternating pairs shall have opposing twist. Where single cover yarns are used, "ZS" twist pattern strand pairs shall be alternated with "SZ" twist pattern strand pairs. Where three ply cover yarns are used, "SZS" twist pattern strand pairs shall be alternated with "ZSZ" twist pattern strand pairs. Each strand shall be composed of either single ply cover yarns or three ply cover yarns. A mixture of strands made up of single ply yarns and strands made up of three ply yarns shall not be permitted in any one rope. The strands may contain polypropylene inside yarns.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T23:09Z
Check the source to verify that this is the current version before use.

3.4.1 Cover yarns. The cover yarns shall be made with polyester fibers covering a polypropylene fiber core. The cover yarns shall contain a minimum of 50 percent polyester fiber when tested as specified in 4.4.5. When three ply cover yarns are used, each of the single yarns of the three plys shall be made with polyester fibers covering a polypropylene fiber core and shall contain a minimum of 50 percent polyester fiber.

3.4.2 Inside yarns, when used. When inside yarns are used in the construction, each strand shall contain a core of polypropylene inside yarns as specified in 3.3.2.2.

3.5 Physical requirements. The finished rope shall conform to the physical properties specified in table I when tested as specified in 4.4.5.

3.5.1 Elongation. The elongation of the rope shall not exceed 35 percent when tested as specified in 4.4.5.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T23:09Z
Check the source to verify that this is the current version before use.

TABLE I. Physical requirements

Circumference (nominal) inches	Cir. tolerance of plus or minus inches	Dia. 1/ (approx) inches	Length 2/ per pound (min) feet	Breaking force (min) pounds	Length per reel (min) feet	Weight 1/ per reel (approx) pounds	Length per 20 double crowns (max) inches
3/4	1/16	1/4	55	1,130	2,250	43	10-7/16
1	1/16	5/16	36	1,710	2,250	66	12-3/4
1-1/8	1/16	3/8	24	2,430	1,600	70	13-13/16
1-1/4	1/16	7/16	18	3,150	1,200	71	15-1/8
1-1/2	1/16	1/2	14	3,960	1,200	91	17-1/2
1-3/4	1/16	9/16	11	4,860	1,200	115	19-5/8
2	1/8	5/8	9.4	5,760	1,200	135	22
2-1/4	1/8	3/4	7.1	7,560	1,200	179	24-5/16
2-1/2	1/8	13/16	5.9	9,180	1,200	215	26-1/2
2-3/4	1/8	7/8	5.0	10,800	1,200	254	29
3	1/8	1	4.0	13,100	1,200	318	31-1/4
3-1/4	3/16	1-1/16	3.5	15,200	1,200	363	33-1/2
3-1/2	3/16	1-1/8	3.1	17,400	1,200	410	35-13/16
3-3/4	1/4	1-1/4	2.7	19,800	600	236	38

1/ Not a specification requirement - included for information purposes only.

2/ A minus tolerance of 10 percent is allowed for dyed rope.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T23:09Z
Check the source to verify that this is the current version before use.

3.6 Finish. No extraneous material shall be added for the purpose of weighing the rope. The extractable matter of the finished rope shall not exceed 4.0 percent when tested as specified in 4.4.5.

3.7 Color. Unless otherwise specified (see 6.2), the color of the rope shall be natural. When colored rope is specified the rope shall be dyed to match an approved standard shade of Olive Drab 7 and both the polyester and polypropylene components of the cover yarns shall be dyed (see 6.3).

3.7.1 Colorfastness. Dyed and finished rope shall show fastness to weathering equal to or better than the standard sample when tested as specified in 4.4.5. When no standard sample has been established, dyed rope shall show good fastness to weathering when tested as specified in 4.4.5.

3.8 Identification. Each reel, spool of rope shall have a ticket (identification tag) or label attached to it for identification purposes. The ticket shall conform to the requirements for type B, grade 15 SU, size 4 or 5 of UU-T-81. The contractor shall use attachment method and color requirement that conform to his commercial practice. All ticket or label entries shall be legibly printed, stamped or typed with insoluble ink. The ticket or label shall contain the following information:

Stock number
Color
Specification number
Length and circumference (size)
Contract number and date
Date of manufacture (Month and Year)
Contractor's name

3.8.1 Identification marker. The manufacturer shall identify his product by inserting a kraft paper or water repellent cotton marker within one strand in all ropes larger than 2 inch circumference. The marker shall be completely enveloped by the cover yarns in the strand in which it is inclosed. The manufacturer's name, year of manufacture and type fiber (Polyester/Polypropylene) shall be clearly printed in block type on the marker. The printing shall not be affected upon exposure to water or mineral oil.

3.9 Put-up. Unless otherwise specified, (see 6.2) the rope shall be furnished without knots or splices on non-returnable reels or spools not larger than 6 feet in diameter in the lengths specified in table I. Pieced lengths will be permitted, but no piece shall be less than 600 feet when measured in the relaxed condition. The ends of all rope shall be cut off squarely and be securely whipped, taped or heat sealed to prevent untwisting or fraying. The reels or spools shall be wound so that each turn and layer is free from entanglement.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T23:09Z
Check the source to verify that this is the current version before use.

3.9.1 Cut lengths. When specified, rope shall be put up in coils in one continuous piece without knots or splices in the minimum length specified (see 6.2). A plus tolerance of 10 percent shall be allowed on the length per coil. The ends of all rope shall be cut off squarely and be securely whipped, taped or heat sealed to prevent untwisting or fraying. The coils shall be so wound that each turn and layer is free from entanglement.

3.10 Moisture content. The moisture content of the rope, as received, shall not exceed 2.0 percent when tested as specified in 4.4.5 (see 6.5).

3.11 Spliceability. The finished rope shall be spliceable and shall not develop yarn displacement or strand cockles in the splicing test specified in 4.4.5.

3.12 Workmanship. The rope shall conform to the quality and grade of product established by this specification. The occurrence of defects shall not exceed the applicable acceptable quality levels established by this specification.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Certificate of compliance. When certificates of compliance are submitted, the Government reserves the right to check test such items to determine the validity of the certification.

4.2 Classification of inspection. The inspection requirements specified herein are classified as follows:

1. First article inspection (see 4.3).
2. Quality conformance inspection (see 4.4).

4.3 First article inspection. When required (see 6.2), the first article submitted in accordance with 3.1 shall be visually inspected for appearance and color. The sample shall be tested for physical and chemical characteristics in accordance with 4.4.5.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T23:09Z
Check the source to verify that this is the current version before use.

4.4 Quality conformance inspection. Except as otherwise specified herein, sampling for inspection shall be performed in accordance with the provisions set forth in MIL-STD-105.

4.4.1 Component and material inspection. In accordance with 4.1 above, components and materials shall be inspected in accordance with all requirements of referenced specifications, drawings and standards unless otherwise excluded, amended, modified or qualified in this specification or applicable purchase documents. In addition, inspection shall be performed for the requirements in table II and a contractor's certificate of compliance thereto shall be furnished and will be accepted for the requirements.

TABLE II.

Characteristics	Paragraph reference
Material	3.3
Single yarn in strand (when applicable)	3.4.1
Plied yarn in strand (when applicable)	3.4.1

4.4.2 End item inspection. The defects listed in table III shall be counted regardless of their proximity to each other. The sample unit shall be one reel, spool or coil as applicable. Ten percent of the length contained on each sample unit, but not less than 100 feet shall be subjected to the visual examination. The lot size for this examination shall be expressed in units of reels, spools or coils each. The acceptable quality level (AQL) shall be 1.5 major defects and 4.0 total defects (major and minor combined) per 100 units. The inspection level shall be level I.

TABLE III. Visual examination defects

Examine	Defect	Classification Major	Minor
Appearance and workmanship	Cut, any	X	
	Chafed or damaged	X	
	Kink, darting yarn 1/, broken or loose end, bulged strand, strand knot, strand or yarn splice	X	

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T23:09Z
Check the source to verify that this is the current version before use.

TABLE III. Visual examination defects (cont'd)

Examine	Defect	Classification Major	Classification Minor
Appearance and workmanship (cont'd)	Ends not cut off squarely		X
	Ends not securely whipped, taped or heat sealed		X
Identification	Omitted, incorrect, or illegible		X
	Insecurely attached or not as specified		X
Color (dyed rope)	Not as specified	X	
	Resists dyeing or spotty dyeing		X
	Specific shade not within established tolerances		X
	Uneven		X
Cleanness	Spot, or stain, clearly noticeable 2/		X

1/ Darting yarn is an internal yarn which projects through the surface of the strand.

2/ At normal inspection distance (approximately 3 feet).

4.4.3 Examination for length and winding. The sample unit for this examination shall be one reel, spool or coil. The inspection level shall be level S-3 and the AQL shall be 4.0 percent defective. The lot size shall be the number of units in the inspection lot. Defects shall be as listed in 4.4.3.1 and 4.4.3.2. The required length of the reel, spool or coil (see table I and 3.9.1) shall be calculated using the following formula:

W x L = Length per reel, spool or coil

Where: W is net weight of the rope in 0.1 pounds

L is the lot average of the weight-length relationship (length per pound) as determined in 4.4.5

4.4.3.1 Length. Defects with regard to length shall be considered to exist if any of the following are determined during inspection:

a. Length of reel, spool or coil less than specified.

Source: https://assist.dla.mil -- Downloaded: 202640-21T23:09Z
Check the source to verify that this is the currentversion before use.

b. Length of unit less than marked on ticket.

c. Rope not in continuous length when specified (see 3.9.1).

d. Any piece on reels or spools less than 600 feet in length.

4.4.3.2. Winding. Defects with regard to winding shall be considered to exist if any of the following are determined during inspection:

a. Improperly or not firmly wound resulting in kinks, knots, entangling or slippage during unwinding or otherwise affecting free unhampered unwinding of rope.

b. Knot or splice or otherwise joining of ends to make a continuous length.

4.4.4 Packaging inspection. Packaging, packing and marking shall be inspected in accordance with the provisions of MIL-C-3131, except that the inspection level shall be S-2 and the AQL shall be 2.5 defects per 100 units.

4.4.5 Testing of the end item. The methods of testing specified in FED-STD-191, wherever applicable, and as listed in table V shall be followed. The physical and chemical values specified in section 3 apply to the average of the determinations made on a sample unit for test purposes as specified in the applicable test methods. The sample size shall be in accordance with table IV. The sample unit for testing shall be one reel, spool or coil of rope. The lot size shall be expressed in units of one reel, spool or coil. All test reports shall contain the individual values utilized in expressing the final result. The lot shall be unacceptable if one or more units fail to meet any requirement. Tests to determine compliance with specification requirements, including quantity of delivery, may be made under prevailing atmospheric conditions. In cases of dispute, tests shall be made upon material which has reached equilibrium under standard conditions as defined in FED-STD-191.

TABLE IV. Sampling for tests

No. of reels, spools or coils in lot	No. of samples	Acceptance No. for each test characteristic	Rejection No. for each test characteristic
Up to 25	2	0	1
26 to 150	3	0	1
151 to 1200	4	0	1
1201 and over	7	1	2

Source: https://assist.dla.mil -- Downloaded: 202610-21T23:09Z
Check the source to verify that this is the current version before use.

TABLE V. Test methods

Characteristic	Requirement reference	Test method
Number of strands	3.4	Visual 1/
Lay of rope	3.4	Visual 1/
Polyester fiber content	3.4.1	4.4.5.1
Circumference	3.5	6003
Length per pound	3.5	6004
Breaking force	3.5	6015 2/
Length per 20 double crowns	3.5	6000 3/
Elongation	3.5.1	6015 2/
Extractable matter	3.6	2611 4/
Color matching (dyed rope only)	3.7	4.4.5.2 1/
Colorfastness to weathering (dyed rope only)	3.7.1	5671 5/
Identification marker	3.8.1	Visual 1/
Moisture content	3.10	2600
Spliceability	3.11	4.4.5.3 1/

1/ One determination per sample unit shall be made and the results shall be reported as pass or fail.

2/ Method 6016 may be used to determine breaking force and elongation for ropes 3 inches in circumference or less.

3/ Except that the length of 20 double crowns shall be determined.

4/ Except that extractable matter shall be determined using the chloroform extraction procedure.

5/ The time of exposure shall be 40 standard fading hours.

Source: https://assist.dla.mil -- Downloaded: 202640-21T23:09Z
Check the source to verify that this is the currentversion before use.

4.4.5.1 Polyester fiber content.

4.4.5.1.1 Procedure. Cut each cover yarn specimen to a minimum length of 10 inches and weigh to the nearest milligram. Carefully separate the polyester multifilament yarn component from the polypropylene monofilament core yarn of the cut specimen by unwinding and weigh the polyester portion separately to the nearest milligram.

4.4.5.1.2 Result. The percentage of polyester fiber shall be based on the following equation:

$$\text{Polyester fiber, percent} = \frac{\text{weight of the polyester component}}{\text{total weight of yarn}} \times 100$$

4.4.5.1.3 Report. Five specimens shall be tested from each sample unit, and the polyester fiber component shall be the average of the results obtained from the specimens tested. The results shall be reported to the nearest whole percent.

4.4.5.2 Determination of shade matching. The color of the dyed rope shall match the standard sample under artificial daylight having a color temperature of 7500° Kelvin and shall be good approximation to the standard sample under incandescent lamplight at 2800° Kelvin (see 6.3).

4.4.5.3 Determination of spliceability. A breaking strength specimen shall be prepared and spliced at each end with four full tucks. The specimen shall then be allowed to rest for 24 hours. Three of the tucks shall then be backed out of the rope. That portion of the rope from which the tucks are removed shall be examined for yarn displacement and strand cockles.

5. PACKAGING

5.1 Preservation-packaging. Preservation-packaging shall be level A or C as specified (see 6.2).

5.1.1 Levels A and C. The rope, put-up as specified, shall be packaged in accordance with the applicable requirements of MIL-C-3131.

5.2 Packing. Packing shall be level A, B or C as specified (see 6.2).

5.2.1 Levels A, B and C packing. The rope shall be packed in accordance with the applicable requirements of MIL-C-3131.

5.3 Marking. In addition to any special markings required by the contract or order, interior packages and shipping containers shall be marked in accordance with MIL-C-3131.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T23:09Z
Check the source to verify that this is the current version before use.

6. NOTES

6.1 Intended use. The rope is intended for general purpose use.

6.2 Ordering data. Procurement documents should specify the following:

(a) Title, number and date of this specification.
(b) Circumference required (see table I).
(c) First article sample is required (see 3.1, 4.3 and 6.4).
(d) Color required, when other than natural (see 3.7).
(e) Put-up and length when other than specified (see 3.9 and 3.9.1).
(f) Selection of applicable levels of packaging and packing (see 5.1 and 5.2).
(g) That polyester/polypropylene rope shall be purchased on a price-per-pound basis net weight.
(h) That purchaser will accept at original weight, any unit which has been shortened or cut for test specimens, if in compliance with this specification.

6.3 Standard sample. For access to the standard sample, address the procuring office issuing the invitation for bids.

6.4 First article. When a first article is required, it shall be inspected and approved under the appropriate provisions of ASPR 7-104.55. The first article should be a preproduction sample. The first article should consist of one reel, spool, or coil of rope. The contracting officer should include specific instructions in all procurement instruments regarding arrangements for inspection and approval of the first article.

6.5 Adjustment for high moisture content. Material furnished containing an excess of moisture will be accepted by an adjustment in weight to the 2 percent moisture basis.

Custodians:

Army - GL
Navy - SH
Air Force - 99

Review activity:

DSA - IS

User activities:

Army - ME, WC

Preparing activity:

Army - GL

Project No. 4020-0255

Source: https://assist.dla.mil -- Downloaded: 202610-21T23:09Z
Check the source to verify that this is the currentversion before use.

STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

OMB Approval
No. 22-R255

INSTRUCTIONS: The purpose of this form is to solicit beneficial comments which will help achieve procurement of suitable products at reasonable cost and minimum delay, or will otherwise enhance use of the document. DoD contractors, government activities, or manufacturers/vendors who are prospective suppliers of the product are invited to submit comments to the government. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements. Attach any pertinent data which may be of use in improving this document. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity.

DOCUMENT IDENTIFIER AND TITLE

Rope, Fibrous, Plaited, Polyester/Polypropylene Dual Fiber MIL-R-43952

NAME OF ORGANIZATION AND ADDRESS	CONTRACT NUMBER
	MATERIAL PROCURED UNDER A ☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?
   A. GIVE PARAGRAPH NUMBER AND WORDING.

   B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY DOCUMENT REQUIREMENT CONSIDERED TOO RIGID

3. IS THE DOCUMENT RESTRICTIVE?
   ☐ YES ☐ NO *(If "Yes", in what way?)*

4. REMARKS

SUBMITTED BY *(Printed or typed name and address - Optional)*	TELEPHONE NO.
	DATE

DD FORM 1 JAN 72 1426 REPLACES EDITION OF 1 JAN 68 WHICH MAY BE USED

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T23:09Z
Check the source to verify that this is the current version before use.